

Q series Patient Monitor Service Manual

GuangDong Biolight Meditch Co.,ltd

No.2 Innovation First Road, Technical Innovation Coast, Hi-tech Zone, Zhuhai , P.R.China

Tel: +86-756-3399900 Fax: +86-756-3399989

<http://www.blt.com.cn>

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Chapter 1 Summary

1.1 About the Manual

1.1.1 Notes

This service manual introduces in detail the hardware composition, assembling, disassembling, testing and trouble-shooting of this product and its relevant spare parts so as to facilitate the maintenance personnel to handle common failures more effectively. This service manual does not provide profound introduction on the product's structure and design mechanism, and in case of unsolvable problems, please contact the after-sale service manual department of our Company.

This service manual introduces the product based on its basic configuration; therefore, part of its content may not apply to the product that you maintain. In case of any queries, please contact the after-sale service department of our Company.

Prior to product maintenance, you are kindly requested to carefully read this service manual, and ensure that you have fully understood its content so as to guarantee that maintenance work can be executed appropriately to avoid causing product or personal damages.

1.1.2 Users of the Manual

This manual is targeted to technicians, maintenance workers of biomedical instruments and after service commissaries.

1.1.3 Intended Use

The monitor is intended to be used for monitoring, displaying, reviewing, storing and alarming of multiple physiological parameters of patients, including ECG, ST segment analysis, arrhythmia analysis, Heart Rate (HR), Respiration Rate (RR), Temperature(Temp), Pulse Oxygen Saturation(SpO₂), Pulse Rate(PR), Non-invasive Blood Pressure(NIBP), Invasive Blood Pressure(IBP), Impedance Cardiograph(ICG), Cardio Output(C.O.), Carbon dioxide(CO₂), Nitrous oxide(N₂O), O₂, anesthetic agents Halothane, Isoflurane, Enflurane, Sevoflurane and Desflurane.

The monitor is intended to be used in healthcare facilities by clinical physicians or appropriate medical staff under the direction of physicians.

Warning:

The monitor is intended for use only by clinical Professionals or under their guidance. It must only be used by persons who have received adequate training in its use. Anyone unauthorized or untrained must not perform any operations on it.

1.2 Safety Information

1.2.1 Terms

The terms in the manual (e.g. danger, warning, caution, and note) point out the hazard, and inform users of the issues they should pay attention to during the operations according to different degrees of severity.

Danger

It indicates urgent danger, which, if not avoided, might cause death, severe personal harms or property loss.

Warning

It indicates potential dangers or unsafe maintenance operations, which, if not avoided, might cause death, severe personal harms, product damage or property loss.

Caution

It indicates unsafe maintenance operations, which, if not avoided, might cause slight personal harms, product damage or property loss.

Note

It emphasizes important notes, provides explanations or interpretation so as to better service this product.

1.2.2 Danger

This Manual does not involve any indicative information of danger class.

1.2.3 Warning

- Only the Company's authorized professional maintenance personnel can disassemble the monitor.
- Prior to the Disassemble the monitor, make sure the AC power is switched off, and batteries are removed so as to avoid high-voltage electric shock.
- Regarding the Disassembled and reassembled parameter modules, before conducting measurements on patients, it is a must to conduct patient leakage current test.

- Please connect the monitor to a socket with protective earth. If the socket does not have protective earth conductor, please do not use the socket and use battery to provide power to the monitor
- Keep the packing materials out of children's reach. Disposal of the packing materials should observe the applicable waste control regulations.

1.2.4 Caution

- When the monitor is conducting some performance tests, make sure the monitor will not be interfered by electromagnetic radiation. Mobile phones, X rays or MRI equipment are all possible interference sources, because they can produce high-intensity electromagnetic radiation.
- Before energizing the monitor, make sure the voltage and frequency of the power conform to the requirements specified in the label on the monitor or in this Manual. During maintenance, pay attention to preventing the monitor from falling, being knocked, or intensely shaken or damaged by other mechanical forces.

1.2.5 Note

For the detailed operation and other information, please refer to the User's manual.

1.3 Equipment Symbols

Symbol	Symbol Note	Symbol	Symbol Note
	Attention: Consult accompanying documents (this manual).	ECG	Short for "Electrocardiogram"
	Non-ionizing radiation	SpO2	Short for "Pulse Oxygen Saturation"
	Dangerous voltage	TEMP	Short for "Temperature"
	Equipotential grounding	IBP	Short for "Invasive Blood Pressure"
	Alternating current	NIBP	Short for "Non-invasive Blood Pressure"

Symbol	Symbol Note	Symbol	Symbol Note
	Auxiliary output connector	CO ₂	Short for “Carbon dioxide”
	DVI display connector	AG	Short for “Anesthetic gas”
	VGA display connector	ICG	Short for “Impedance cardiography”
	USB socket	C.O.	Short for “Cardio Output”
	Network connector		Defibrillator synchronization output connector
	Manufacture date		Auxiliary plug-in box connector
SN	Serial number		Manufacturer
	CE mark	REF	Catalog number
	Symbol for the marking of electrical and electronics devices according to Directive 2002/96/EC.		
	Type CF applied part, defibrillation protected The unit displaying this symbol contains an F-Type isolated (floating) applied part providing a high degree of protection against shock, and is defibrillator-proof.		
	Type BF applied part, defibrillation protected The unit displaying this symbol contains an F-Type isolated (floating) applied part providing a high degree of protection against shock, and is defibrillator-proof.		

Chapter 2 Monitor Introduction

2.1 Main Frame

2.1.1 General Introduction

The monitor is intended to be used for monitoring of multiple physiological parameters of patients, including ECG, ST segment analysis, arrhythmia analysis, Heart Rate (HR), Respiration Rate (RR), Temperature(Temp), Pulse Oxygen Saturation(SpO₂), Pulse Rate(PR), Non-invasive Blood Pressure(NIBP), Invasive Blood Pressure(IBP), Impedance Cardiograph(ICG), Carbon dioxide(CO₂), Cardio Output (C.O.), Anesthetic gas(AG), Oxygen(O₂) and so on.

Q7:



Q5:



Q3:



This monitor also provides the following functions:

- ◆ Conduct visual and audio alarms on patients' physiological abnormalities and equipment's technical problems.
- ◆ Conduct real-time display, review, storage, and output of the monitored parameters or waveforms.
- ◆ Multiple operation input mode, including keys, trim knob, touch-screen, keyboard and mouse.
- ◆ Wire connection with CMS system provided.
- ◆ Program upgrade.

2.1.2 Appearance, Connector, Slot

Q7 :



Q5、 Q3:



2.1.3 Hardware Structure

The host monitor is composed of the following parts:

- ◆ Input system: key board, trim knob, touch-screen, power switch
- ◆ Output system: display screen, alarm indicator board, recorder, loudspeaker and SD card (optional)
- ◆ Processing and communication system: main-board, based board for main board and host plug-in box
- ◆ Power management system: battery, battery interface board and power module
- ◆ Equipment interface system: SOPA board(USB,VGA,RJ45), SOPB board (BNC,RJ11)

2.1.3.1 Input System

- ◆ Key board

Q3/Q5: On the right side of the screen, 5 key inputs are provided, and connected to the key board.

Q7: On the bottom of the screen, left and right keys are provided, and connected to the key board.

- ◆ Trim knob

The trim knob coder supports rotations in clockwise, counter-clockwise directions and press operation.

- ◆ Touch-screen

The touch-screen supports touch control function, and it can be calibrated and connected with the key board.

- ◆ Power switch and indicator board

The power switch and the power for LED indicator panel for the host monitor. It has three groups of LEDs used for indicating the status of AC power, battery and the host monitor. It is connected with the key board.

2.1.3.2 Output System

- ◆ Display

The host monitor has a built-in high-brightness LCD display.

Q3/Q5: The display is connected with the key board and the backlight board, and the DC power it uses for operation is provided by the main-board and is supplied to the backlight board by the key board and then supplied to the display by the backlight board.

Q7: The display is connected with the key board, and the DC power for operation is provided by the main-board and is supplied to the display by the key board.

◆ Alarm light

The host monitor comprises two kinds of alarm lights: physiological alarm lights and technical alarm lights. The physiological alarm lights can emit light in two colors of yellow and red and technical alarm lights can emit light in blue color. They both support different flashing frequencies.

◆ Recorder

The recorder receives the data transmitted by the main-board and then transmits the data to the thermal print head for print-out. The front panel of the recorder has an orange indicator (it lights up when an alarm or error occurs) and a green indicator (it lights up during normal operation). It is connected with the main-board.

◆ Loudspeaker

The loudspeaker is used for sounding alarm, key tones, heartbeat tones and pulse tones and supports the multi-volume function. It is connected with the main-board and its DC power is supplied by the main-board.

2.1.3.3 Processing and Communication System

◆ Main-board

As the heart of the monitor, the main-board accomplishes such a series of tasks as input and output control, data storage and processing, display processing, system control, communications management, printing management and failure alarm.

◆ Based board for main board

It is connected to the main-board directly, and is responsible for the connection and relevant communications control with other components and peripheral devices. It has the following main interfaces:

- LCD: connect with built-in display
- Key board: the backlight, the touch screen and key
- Infrared interface board: plug-in module
- BM100A module: connected to the multi-parameter module
- SOPA interface board: display output VGA/DVI
- SOPB interface board: USB, +RS422
- Loudspeaker: connect with the loudspeaker
- SD card: connected to the SD card:
- Power module: connect with the power module
- Battery interface board: battery
- Recorder: connect with the recorder

2.1.3.4 Power Management System: Battery, Power Module, Battery Interface

◆ Battery

the host monitor supports DC power supply, and maximum of two rechargeable lithium batteries of 11.1V/4.0Ah can be used at the same time , the Q7 working hour is as long as 3 hours, and Q3/Q5 the working hour is as long as 4 hours.

◆ Power module:

AC: 100-240V, 50/60Hz, ≤150VA;

◆ Battery interface board:

Connect to the battery

2.1.3.5 Equipment Interface System:

SOPA Interface Board, SOPB Interface Board



SOPA interface board:

1. VGA interfaces: It can be connected with DVI or VGA video output.
2. Network interface: It is networked with the central monitoring system via the internet;
3. USB interface: It can connect with mouse and keyboard to facilitate the operation and entry of relevant information; Software update, supply the power to WiFi module and so on.

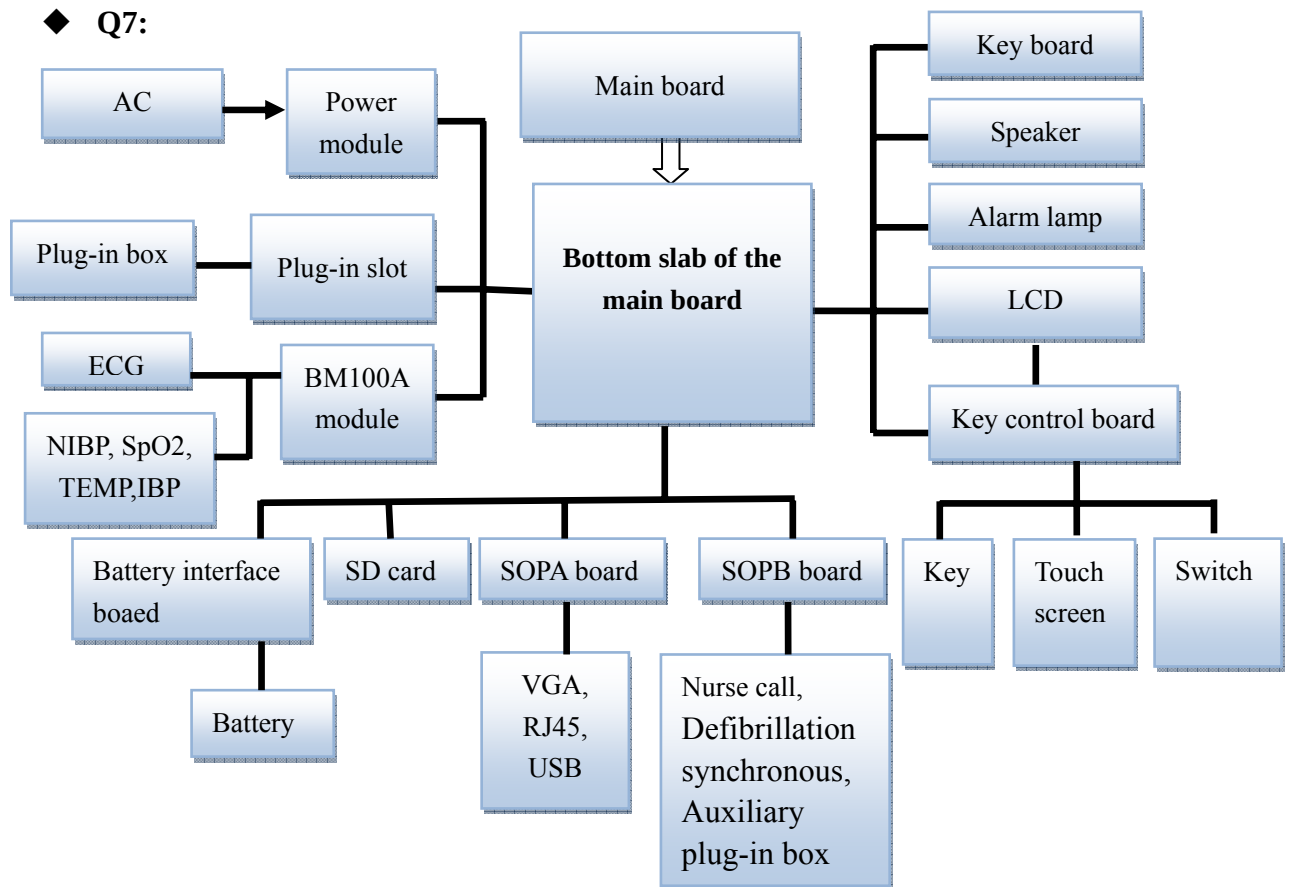
SOPB interface board:

1. Defibrillation synchronous output/Analog output interface, it is connected with the defibrillation device and sends defibrillation synchronous signals. And also connected to device, such as oscilloscope and sends analog signals.
2. Nurse call output interface, It can be connected with the hospital's call system, which, in case of an alarm, sends out a nurse call signal to alert the nurses.
3. Auxiliary plug-in box interface: Connect to auxiliary plug-in box, to expend plug-in modules.

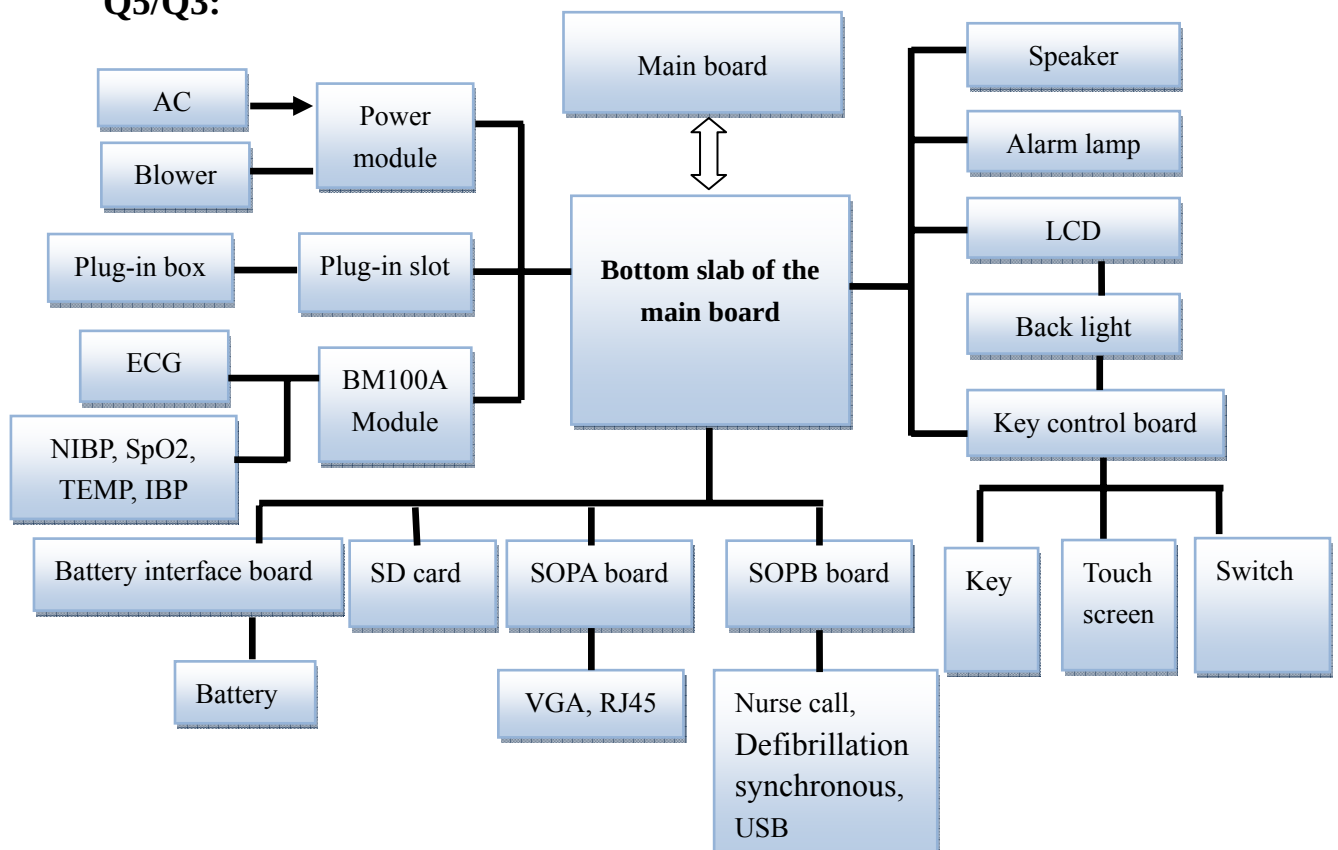
2.1.4 Design Principle

Schematic diagram:

◆ Q7:



Q5/Q3:



2.2 Other Parameter Modules

IBP module, TEMP module, Nellcor SpO2 module, Masimo SpO2 module, CO2 module (contains Mainstream CO2, Side stream CO2 and Micro stream CO2), AG module, ICG module and C.O. module.



2-IBP module



Nellcor SpO2 module



2-TEMP module



Sidestream CO2 module



Microstream CO2 module



AG module



ICG module



C.O. module

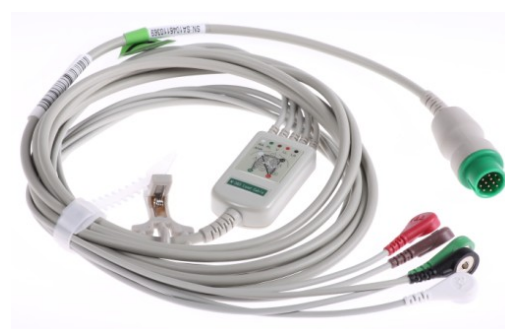
2.3 Accessories

2.3.1 General Accessories

General accessories mostly contains ECG cable, SpO2 sensor, NIBP cuff, Temperature probe, in addition, it contains mainstream CO2 measurement groupware, micro stream CO2 measurement groupware, AG measurement groupware, ICG measurement groupware, C.O. cable and pulmonary artery catheter. More than this, the monitor also could install ledge, cart, and rechargeable lithium ion battery of host monitor.



12PIN 3-lead ECG cable



12PIN 5-lead ECG cable



12PIN 12-lead ECG cable



SpO2 cable



BLT 9PIN adult SpO2 sensor



BLT 9PIN SpO2 sensor



BLT 9PIN Neonatus SpO2 sensor



Nellcor SpO2 cable



Nellcor adult SpO2 sensor



MasimoSpO2 cable



Masimo adult SpO2 sensor



NIBP cuff



TPU air tube



Temperature body surface probe



Body-cavity probe



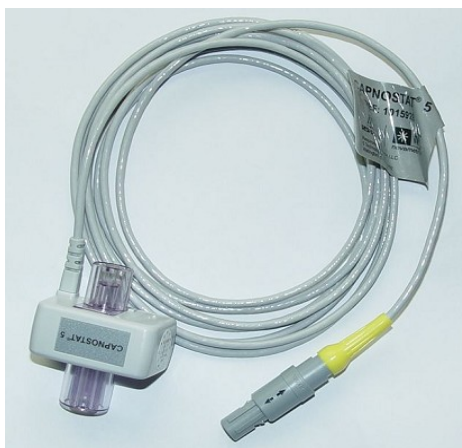
IBP cable (UTAH)



IBP cable (BD)



4 PIN change to 6PIN IBP cable



Main stream CO₂ sensor and adapter (C5)



Microstream CO₂ sensor (LoFlo)



Mainstream AG sensor (IRMA AX+)



Sidestream AG sensor (ISA OR+)



ICG lead wire



ICG cable



C.O. cable



Temperature probe



Pulmonary artery catheter



Wall Mount



Cart



Rechargeable lithium ion battery

2.3.2 Management and Maintenance of Accessories

Refer to the User's manual.

Chapter 3 Testing

3.1 Brief Introduction

To ensure the long-term and stable working of the monitor, maintenance personnel need to conduct regular checks, maintenance and testing on it. This chapter provides the basic testing methods of the monitor and recommends the applicable testing frequency and testing tools, and the maintenance personnel are requested to conduct maintenance and testing on the monitor according to actual demands and choose applicable testing tools.

The testing and verification methods provided by this chapter are mainly used for verifying whether the monitor's performance has reached the specification requirements. During the tests, if the test results can not reach the requirements, it indicates the monitor or a certain function module of the monitor has failures and they need to be repaired or replaced. If you have any query, please contact the after-sale service department of our Company.

Caution:

- **All the testing must be done by qualified service personnel.**
- **When setting up and changing the contents in the two menus of 【Config Manage】 and 【Machine Maintenance】 , please be cautious, otherwise it might cause data loss.**
- **Prior to executing tests, the service personnel are kindly requested to ensure the applicability of the testing tools and cables and service personnel shall be familiar with the use of these testing tools.**

3.2 General Testing

3.2.1 Visual Check

Visual check is mainly for overall appearance check on the monitor, and if the monitor has no obvious physical damage, visual check is qualified.

Check items are as follows:

1. Whether the host monitors' enclosure, display screen or keys have any physical damage.
2. Whether each module have any physical damage.

3. Whether external cable has any wear or tear, whether the connector pins have any loosening or distortion.
4. Whether the peripheral interface of the monitor has any loosening or pin distortion.
5. Whether the safety labels or name plates are legible.

3.2.2 Startup Testing

Startup test is used to verify whether the monitor can be normally started to work, and if the monitor can accomplish startup according to the following procedures, the startup test is qualified. The procedures are as below:

1. Connect the monitor to the AC power. AC power indicator lights up.
2. Press the power switch, the technical alarm indicator lights up in blue, and the physiological alarm indicator lights up in yellow and red in turn and then goes off, the screen displays the startup image.
3. After the startup image disappears, the system produces a “dang” sound and goes into the main screen; at the same time, the technical alarm light goes off.
4. The monitor goes into the system’s main screen and normal startup is completed.

Warning:

If the monitor is mechanically damaged, or if it is not working properly, do not use it for any monitoring procedure on a patient. Contact your service personnel.

Caution:

- **The monitor does not have mains switch. The monitor is switched completely only by unplugging the power cable from the AC power source. In order to unplugging and plugging it, the socket on the wall must approximate to monitor.**
- **For measurements in or near the heart we recommend connecting the monitor to the equipotential grounding system. Use the green/yellow equipotential grounding cable and connect it to the terminal labeled with the symbol .**

3.3 Performance Testing

3.3.1 ECG Testing

Testing tool: ECG Simulator

1. Use the ECG lead wire to connect simulator and the monitor.
2. The setting of the simulator is as follows: HR 80 bmp, gain selection is 10mm/mV.

3. Observe whether the monitor displays correct ECG waveforms and HR.
4. Adjust the ECG gains and HR values and verify whether the monitor's ECG waveforms and HR values have relevant changes.

3.3.2 NIBP Testing

Testing tool: BP simulator

1. Insert the cuff extension tube plug into the NIBP socket of monitor and make the plug and socket in good contact.
2. Connect the other side of the cuff extension tube with the simulator, and the simulator's configuration: preset: 120/80 (93), cuff: internal Adult.
3. Measure blood pressure; observe whether there is a big deviation between the measured value and the set value of the simulator. (generally within 3mmHg)

3.3.3 SpO2 Testing

Testing tool: SpO2 simulator

1. Plug the SpO2 extension cable tightly into the SpO2 socket of the monitor.
2. Connect one side of SpO2 probe to the SpO2 extension cable, and connect the other side with the SpO2 simulator.
3. The parameter setup in the simulator is as follows: SpO2 is 96%; pulse rate is 80 bpm.
4. Observe whether the monitor displays correct SpO2 waveforms and pulse rate.

3.3.4 TEMP Testing

Testing tool: resistance box

1. Connect the two sockets of any of TEMP socket of monitor with the two sides of the resistance box using two wires.
2. The setup of resistance box is 1354.9Ω (the corresponding TEMP value is 37°C).
3. Observe the monitor's display value shall not exceed $37\pm0.2^{\circ}\text{C}$

3.3.5 Resp Testing

Testing tool: ECG simulator

1. Connect the simulator with the monitor using the ECG lead wire.
2. The setup of the simulator: base resistance is 1000Ω ; variable resistance is 1Ω ; Resp rate is 30 rpm.
3. Observe whether the Resp waveform displayed by the monitor is normal. Resp value shall not exceed 30 ± 2 rpm.

3.3.6 IBP Testing

Testing tool: IBP simulator

1. Connect the simulator with the IBP socket of the monitor.
2. Set the simulator to zero pressure.
3. Press the “zero” key of IBP module to conduct zero calibration of the parameter module.
4. Set the monitor’s static pressure $p = 200\text{mmHg}$.
5. The display value of the monitor shall not exceed $200 \pm 4\text{mmHg}$.
6. In case the error exceeds $\pm 4\text{mmHg}$, conduct calibration on the IBP parameter module.
7. Set up the simulator IBP as 120/80, 120/25, 120/0, 25/0, and observe whether the waveforms and data displayed by the monitor are correct.

3.3.7 Recorder Testing

1. Print the ECG waveform, the recorder shall print normally and the printing is clear and consistent.
2. In case of such failures as paper lack or paper jam, the screen shall have relevant indication and after restoration, it shall work normally.
3. Conduct the alarm printing of various parameters, turn on the alarm record switches of the various parameters, set up different alarm limits, and in case of a parameter alarm, there shall be the relevant printout of the parameter alarm.

3.3.8 Mainstream CO₂/Microstream CO₂ Testing

1. After plugging the module into the plug-in box, observe whether the module indicator lights up, and then observe whether the host monitor’s screen displays “CO₂ module inserted”, and after entering **【Module Config】**, observe whether the CO₂ module icon appears.
2. Connect the sensor, put the sensor in front of the mouth and breathe, observe whether CO₂ waveforms and values appear on the screen, and then put the sensor still in the air, and in 10s, the alarm info of “CO₂ RESP suffocation” is generated.
3. Remove the sensor, and the screen displays “CO₂ sensor off”.

3.3.9 Sidestream CO₂ Testing

1. Plug the module into the plug-in box, observe whether the module indicator lights up, and then observe whether the host monitor’s screen displays “CO₂ module

inserted ”, enter **【Module Config】** and observe whether the CO2 module icon appears.

2. Listen to the module and check whether there is a sound of motor rotation and the CO2 scanning lines appear on the screen.
3. Remove the module, the module motor sound stops, and the relevant data of CO2 on the host monitor's screen disappear.

3.3.10 AG Testing

1. Plug the module into the plug-in box, observe whether the module indicator lights up and then observe whether the host monitor's screen displays “AG module inserted”, enter **【Module Config】** and observe whether the AG module icon appears.
2. Connect the sensor, and observe whether the screen of the host monitor displays the scanning lines and then the AG parameter zone.
3. Remove the module, and the screen displays “AG sensor off”.

3.3.11 ICG Testing

Testing tool: ICG simulator

1. Use standard ICG cable and sensor, and configure the ICG simulator and the monitor into a testing system.
2. Select **【ICG Set】**→**【Input info>>】**, and set up the relevant parameters as follows:

Gender	Male	Sys	120 mmHg
Age	30 years	Dia	70 mmHg
Height	170cm	MAP	86 mmHg
Weight	65kg	CVP	6 mmHg
		PaOP	16 mmHg

3. Turn on the simulator and set it up as Mode 1 and start up ICG monitoring.
4. After the monitor's measurement value is stabilized, record the measured data.
5. Switch the simulator to Mode 2, start up ICG monitoring, and after the monitor's measurement value is stabilized, record the measured data again.
6. Verify the data measured by the monitor comply with the requirements of the simulator setup.

3.3.12 C.O. Testing

Testing tool: FLUKE vital sign simulator

1. Plug the module into the plug-in box, observe whether the module indicator lights

- up and then observe whether the host monitor's screen displays "C.O. module inserted", enter **【Module Config】** and observe whether the C.O. module icon appears.
2. Connect one terminal of the "C.O. cable" to the C.O. module, the other terminal connect to the vital sign simulator, then select the simulator, enter 2.5L/min of the cardiac output wave, 24°C of injection and 37°C of base temperature of blood;
 3. Monitor select the C.O. parameter zone to enter into the menu of **【C.O. Setup】**, and select the **【TI Source】** to manual, select the **【Setup TI】** to 24°C, select **【Cath. Const】** to 0.595, then close the **【C.O. Setup】** menu.
 4. Select **【C.O. Measure】**, to open the C.O. measurement window, when the screen show that "Ready for new", select **【Start】** to measure the C.O., when the screen show that "Injection now ...", enter the start key of the simulator, observe the wave, waiting for the measured values.
 5. Change the C.O. wave to 5 L/min, 10 L/min, then measure again and record the measured values.
 6. The measured values compare to the value entered on simulator, the standard error shall be $\pm 0.2\text{L/m}$.

Chapter 4 Maintenance

Enter operation mode, select **【Maintenance】** →input password:785623,and then ,return main screen, select **【Maintenance】** ,and enter maintenance screen:

Machine Maintenance	Factory Maintenance	Language Select	Software Upgrade	Setup	System Alarm Setup	Nurse Call	BNC port	Defibrillator Synchronization
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4.1 Maintenance of Monitor

ECG Signal	Network Set	NIBP Maintenance	Phasein Maintenance	IBP Pressure Calibration	CO2 Module Maintenance	Data Statistics
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4.1.1 ECG Signal

It contains standard ECG signal and normal ECG signal.

4.1.2 Network Setting

Set IP address of monitor. Setting IP address and device number on the menu when it connects central monitoring system.

4.1.3 NIBP Maintenance

Based on different persons (adult, pediatric, neonate and Hyper Mode) to maintenance on the menu:

Manometer Test	Over-press Test	Soft overpress Test	Air leakage Test	NIBP Reset	NIBP Calibration (100mmHg)
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4.1.3.1 Manometer Test

It is used for factory maintenance.

4.1.3.2 Over-press Test

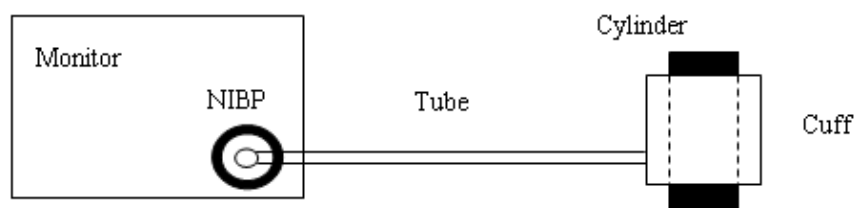
It is used for factory maintenance.

4.1.3.3 Soft overpress Test

It is used for factory maintenance.

4.1.3.4 Air leakage Testing

1. Plug the air pipe plug of cuff into the connector (NIBP) of monitor until the plug and socket contact well
2. Tie the cuff to a metal cylinder.
3. Select **【Maintenance】** → input password → **【Main Menu】** → **【Maintenance】** → **【Machine Maintenance】** → **【NIBP Maintenance】** → **【Adult】** → **【Air leakage Test】**, Then the prompt “Air Leakage Test” will appear on the NIBP parameter area indicating that the system has started performing Air Leakage test.
4. The system will automatically inflate the pneumatic system to about 180mmHg.
5. After 20 seconds or so, the system will automatically open the deflating valve, which marks the completion of an air leakage test.
6. If no error information displays on NIBP parameter area, it indicates that the airway is in good situation and no air leaks exist. However if the prompt “AIR SYSTEM LEAK” appears in the place, it indicates that the airway may have air leaks. In this case, the user should check for loose connection. After confirming secure connections, the user should re-perform the air leakage test.
7. It will stop air leakage test when you press the stop key on the board.



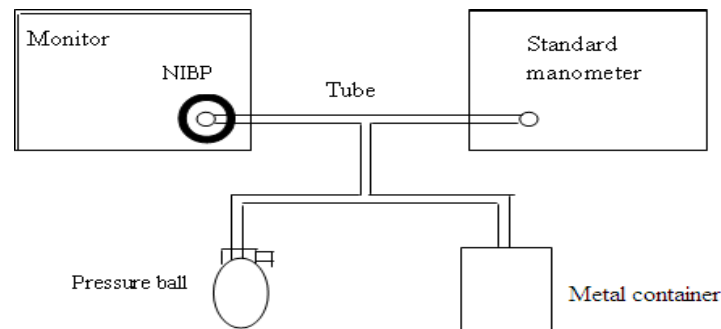
4.1.3.5 NIBP Resetting

NIBP module reset.

4.1.3.6 NIBP Calibration

Calibration tools: 3 way connector, pipe, roundness pump, metal container (500±25 ml), standard manometer (Calibration already, precision over 1 mmHg)

1. Connect monitor, manometer, roundness pump and metal container as follows.



2. Reading of manometer should be 0 before deflate, if not, cut the connection until it return to zero.
3. Select **【Maintenance】** →input password→ **【Main Menu】** → **【Maintenance】** → **【Machine Maintenance】** → **【NIBP Maintenance】** → **【Adult】** → **【NIBP Calibration】** .
4. Turn up pump output pressure to 150 mmHg,the pressure showed by monitor and consult manometer can't be over 3 mmHg,if not, setting**【Press Calibration】**for 150 mmHg,select **【Ok】** key in right of the menu.

4. 1. 4 PHASEIN Maintenance

CO2 Calibrate	N2O Calibrate	O2 Calibrate	AA Calibrate	IR/O2 Delay	Pump	O2 Option	O2 Sensor Type
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This menu is mainly used for maintenance AG module of PHASEIN.

For precision of the gas module testing, we suggest zeroing the gas module regularly.

We need put up Calibration to gas module as follows:

- The measured reading occurs error;
- A“Zero Required” alarm message is displayed;
- Airway adapter is replaced.

As following procedures:

1. Snap a new AG airway adapter onto the AG sensor. Ensure that the airway adapter is not connected to the breath circuit of patient. The presence of ambient air (21% O₂ and 0% CO₂) in the AG airway adapter is very important.
2. Select **【Zero】** in the AG parameter setting menu, this will start zeroing. The monitor zeroes the module and displays the“Zero” message for about 5s.
3. There will be a blinking green light on the AG sensor LED during 5s while Zero Reference calibration is in progress. Wait until the AG indicating lamp light on green

and the reading of oxygen on the screen is “21”. (When using CO₂ module, the reading of CO₂ on the screen is “0”.) The message disappears upon completion of the zeroing.

Caution:

User may only perform zero reference calibration under the instruction of the technical personnel authorized by manufacturer.

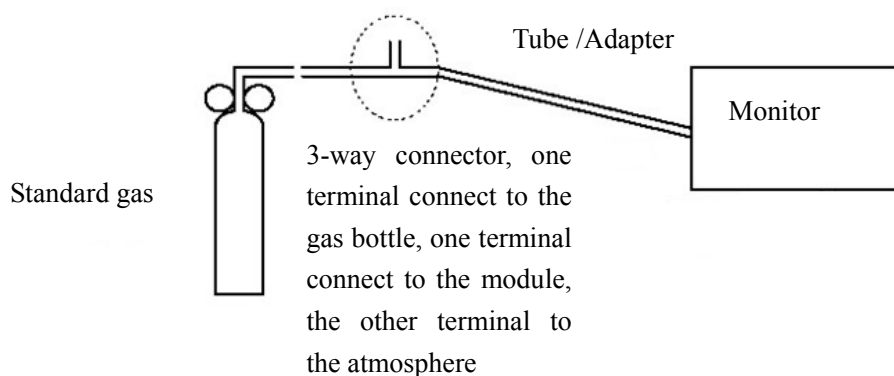
Warning:

Incorrect zero reference calibration will result in false gas readings.

Gas calibrate:

1. Calibrate tools: Standard gas, gas tube, 3-way connector

2. Connecting graph:



2. Operating steps:

Connect the air outlet tubes of standard gas bottle to one terminal of the 3-way connector, and the middle terminal of the 3-way connector opens to the atmosphere; connect the gas sampling tube provided by our company to the remaining terminal of the 3-way connector (see the circle part of the above figure), connect the other terminal of the sampling tube to the probe, the other side of the probe connect to the monitor.

After the airway is connected, according to the concentration of the standard gas, into the gas calibrate menu (CO₂ Calibrate, N₂O Calibrate, O₂ Calibrate, AA Calibrate), and enter relevant concentration value, then press **【OK】**. Observe whether the value displayed on the screen is consistent with the concentration of the standard gas, and the error shall be ± 0.1 , and the observation time shall be no less than 1 minute.

4.1.5 IBP Pressure Calibration

The menu is mainly for pressure (IBP) calibration and factory maintenance.

4.1.6 CO2 Module Maintenance

The menu is mainly for maintenance of the sidestream CO2 module.

Offset Calibrate	Gain Calibrate	CO2 Wave Value	Wave Fill Type
------------------	----------------	----------------	----------------

◆ Calibration of Sidestream CO2 module

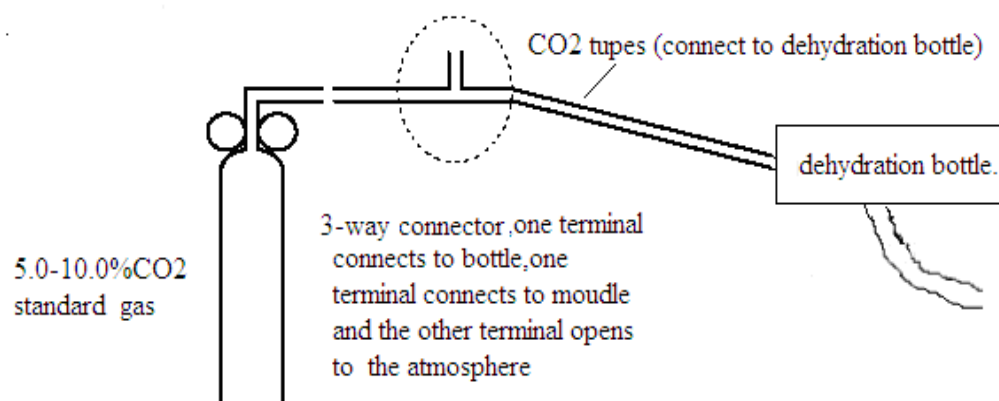
Ex-factory calibration had been conducted on this monitor and under general conditions (with the exceptions of the following three situations), the user can directly conduct measurement. In the presence of the following three situations, please conduct gain calibration on the CO2 module;

- The CO2 module has been used for half a year to one year;
- Clinicians doubt the accuracy of the readings;
- Since last calibration, there have been substantial changes in atmospheric pressure or altitude.

1. Preparation of calibration instruments:

CO2 standard gas (with a concentration of about 5%, and a variety of concentrations can also be prepared), 3-way connector (used for connecting the gas bottles, CO2 tubes and open to air. The 3-way connector mentioned here is not the one in the appendix when our Company sells the modules, and the 3-way connector in the appendix is used for connecting the respirator or the anesthesia machine), dehydration bottle and CO2 tubes.

3. Make connections in accordance with the following figure:



A. Connect the air outlet tubes of standard gas bottle to one terminal of the 3-way connector, and the middle terminal of the 3-way connector opens to the atmosphere; connect the gas sampling tube provided by our Company to the remaining terminal of

the 3-way connector (see the circle part of the above figure), connect the other terminal of the sampling tube to the dehydration bottle and the dehydration bottle is inserted into the bracket of the dehydration bottle.

B. After the airway is connected, start the sampling pump (i.e. start up the CO₂ module), warm up the monitor for 30 minutes, and during the warming up, it is allowed to press the sampling tube with a hand for a short time, and as a result, the pump on the module will increase in speed quickly and it can be heard that the sound is markedly increased; in case the pump does not speed up when the sampling tube is pressed by hand, or the pump has only small increase in speed-up, it is due to the improper connections of the dehydration bottle and the bracket and there is a leakage, or at a certain location of the tube from the bracket to the module there is phenomenon of air leakage, at this time, it is inappropriate to conduct calibration, and it is necessary to check out the leaking point and eliminate the leaking point before conducting calibration, otherwise, due to air leakage, it will lead to the lowering of the measured values. After monitor warm-up, switch on the standard gas, observe the rotation speed of the pump on the module; if the pump rotates very slowly, it indicates the air flow of the standard gas output is too big, leading to the fact that high pressure is exerted on the module. At this time, it is necessary to adjust the volume of the standard gas output, until the pump on the module rotates at a normal speed.

3. Gain calibration:

A. Turn the trim knob, and after moving the cursor to the CO₂ menu, press the key and change the unit to “%”;

B. Turn the trim knob, and after moving the cursor to the main menu, press the key to enter **【Main Menu】** → **【Maintenance】** → **【Machine Maintenance】** → **【CO₂ Module Calibration】**, select **【Offset Calibration】** as “manual”;

C. Enter CO₂ menu, select **【Zero】**, and enter relevant value according to the concentration of the CO₂ standard gas and then press **【Ok】**;

D. Observe whether the value displayed by “FiCO₂” is consistent with the concentration of the standard gas, and the error shall be ± 0.1 , and the observation time shall be no less than 1 minute.

Attention:

Within one or two minutes after the standard gas passes through the module, it is necessary to enter the calibration value, and in case the standard gas passes for a long time (for example exceeding 5 minutes) without entering the calibration

value, the display value will be slightly lower.

E. Stop sampling pump;

F. Turn the trim knob and after moving the cursor to the CO₂ menu, press the key, change the popped-up **【Offset Calibration】** to “Automation”, and the unit is set as “mmHg”.

◆ CO₂ Wave value: Hide or show the CO₂ wave value

◆ Wave Fill Type: Select **【Normal】** or **【Fill】** type of the Wave Fill way.

◆ Zeroing main stream/micro stream CO₂ module

Ensure that the nasal cannula or airway adapter is not connected to the patient or close to any source of CO₂ (including the patient's, your own, exhaled breath and ventilator exhaust valves).

Select **【Zero】** in the CO₂ parameter setting menu, this will start zeroing. The monitor zeroes the sensor and displays the message “Zeroing” for about 15-20s on the screen. The message disappears upon completion of the zeroing.

Caution:

- Always ensure that the sampling tube is properly connected to the micro stream CO₂ sensor before zeroing.
- Always ensure that the mainstream CO₂ sensor is properly connected to the airway adapter before zeroing.
- Do not attempt zeroing for 20s after removing the adapter or cannula from the patient's airway. This time allows any CO₂ remaining in the adapter or cannula to dissipate before zeroing.
- Do not attempt to zero the sensor while the adapter or cannula is in the patient's airway.
- Do not attempt zeroing if the temperature is not stable.
- Zeroing with CO₂ in the adapter or cannula can lead to inaccurate measurements or other error conditions. If you attempt zeroing while CO₂ remains in the adapter or cannula, the time required to zero the sensor may be increased.

4.1.7 Module Information

Data Statistics	Module Status	Channel Module Info.
-----------------	---------------	----------------------

To do the data statistics for each parameter, each module channel, and use for factory debug.

4.2 Factory Maintenance

4.2.1 Factory Setting

Setting parts configurations of monitor, it contains:

ECG Lead Type: 5 Lead/12 Lead

SPI slot: With/NULL

Printer: With/NULL

Relay Type: Normal Open/ Normal Close

Screen Size: 15/12.1/10.4

Touch Scr.: With/NULL

ECG 12: With/NULL

IBP: With/NULL

Scanner: With/NULL

HL7: On/Off

BIS: On/Off

CSM: On/Off

ICG: On/Off

Rainbow: On/Off

NIBP: On/Off

4.2.2 Language Setting

Select the languages users needed, including Chinese, English, Spanish, Italian, Polish, Czech, German, Russian, French, Turkish, Greek, Hungarian, Portuguese and so on.

4.3 Language Selecting

Select the languages users needed.

4.4 Software Upgrading

Software of this monitor can be upgraded through USB disk and the Internet, and the monitor's logo can be changed through the Internet.

Caution:

Please confirm the device was connected to the AC power before the upgrading, confirm the inside battery was full charged. The device out of power may leading the device unable to starting when at the upgraded process, then

must send back to factory for maintain.

4.4.1 USB Disk Upgrading

USB disk upgrading can only upgrade the monitor's operation software, and the procedure is as follows:

1. Store the monitor's software to be installed into the USB disk. **Attention: file name cannot be changed in anyway.**
2. Insert the USB disk into the USB socket on the back interface plate of the monitor.
3. After inserting the USB disk, the monitor will automatically read the info of the USB disk and the monitor will automatically upgrade the software.
4. When the monitor's screen displays "Upgrade completed, please unplug USB disk", unplug the USB disk and then restart the monitor.

4.4.2 Network Upgrading and Change of the Logo of the Monitor

Needed tool: a cross network cable or a switch and two common network cables, TFTP server software, and the concrete method is as follows:

1. Run TFTP software.
2. Change the name of the file to be downloaded as bse-logo and put it into the TFTP download catalog.
3. Connect the network to enter the menu of software upgrading, set up correct IP address (keep it consistent with the PC's IP address), and then select "LOGO" for the upgrading type.
4. Upon completion of the upgrading, restart the monitor.

Attention:

The file name of the downloaded startup image is bse-logo.

4.5 Entire Setup

Set the whole configures of the monitor, include:

AC Filter Freq.: 50Hz/60Hz

Quick Admit Pat.: Name/MRN

4.6 System Alarm Setup

Set volume, volume interval, and category of the alarm.

4.7 Nurse Call Type

Select the **【Continual on】** or **【1s Pulse】** .

4.8 BNC Port

Select **【Analog Output】** or **【Defib. Sync.】**

Chapter 5 Troubleshooting

5.1 Overview

This chapter conducts categorization based on the failure type of the monitor, and for troubleshooting, please refer to the failure table and conduct checks according to the sequence.

The recommended solutions given in this chapter can solve most of the equipment failures, but may not necessarily cover all possible failures. In case of any failures not included in this chapter, please contact the after-sale service department of our Company.

5.2 Common Failure Analysis and Trouble-shooting

Phenomenon	Possible reason	Trouble-shooting
Monitor can be turned on, but LED or battery LED does not light up	AC power is not connected, or battery capacity is insufficient	Check whether the AC power is connected correctly, or whether the battery capacity is sufficient.
	Cable failure	1. Check whether the power switch and the cable from the indicator to the key board, the cable from key board to the main control unit and the cable from the power module to the main control unit are properly connected. 2. Check whether the cables and plugs are damaged.
	Power module is damaged	Replace the power module.
Battery cannot be recharged	Power module is damaged	Replace the power module.
	Battery failure	Replace the battery.

Black screen, monitor runs normally	Cable failure	1. Check whether the cable from the LCD display screen to the main-board is properly connected. 2. Check whether cables and connectors are damaged.
	Backlight board is damaged	Replace the backlight board.
	LCD display screen is damaged	Replace the LCD display screen.
Keys have no response	Cable failure	1. Check whether the cable from the key board to the main-board is properly connected. 2. Check whether cables and plugs are damaged.
	Key board is damaged	Replace the key board.
Trim knob has no response	Cable failure	1. Check whether the cable from the knob to the key board and the cable from the key board to the main-board are properly connected. 2. Check whether cables and plugs are damaged.
	Trim knob is broken	Replace the trim knob coder.
Recorder is unable to print	Recorder module is prohibited	1. Check whether the recorder indicator lights up. 2. If the indicator lights up, recover the function of the recorder module in the menu of 【Factory Maintenance】 . If the indicator does not light up, it indicates that there are other failures.
	Printing paper is mounted reversely	Install the printing paper again.

Recorder is unable to print	Cable failure	1. Check whether the cable from the recorder to the main-board is properly connected. 2. Check whether cables and plugs are damaged.
	Recorder failure	Replace the recorder.
Analog signal/nurse call have no output	Relevant output function is not turned on	1. Under the maintenance mode, 【Main menu】 → 【Maintenance】 → 【Factory Maintenance】 , selection type is 【Analog output】 or 【Nurse call】 .
	Cable is loosened	1. Check whether the cable from the SOPB interface board to the main-board is properly connected. 2. Check whether cables and plugs are damaged.
	DVI interface board is damaged	Replace DVI interface board.
USB equipment can not be used (peripheral equipment has no damage)	Cable failure.	1. Check whether the cable from the SOPB interface board to the main-board is properly connected. 2. Check whether cables and plugs are damaged.
	USB interface board is damaged	Replace the USB interface board.
	Main-board is damaged	Replace the main-board.

Chapter 6 Maintenance and Disassembling

6.1 Tool

During the process of disassembling and replacing the parts, you may need to use the following tools:

1. Cross screwdriver
2. Slotted screwdriver
3. Socket wrench
4. Sharp-nose plier
5. Bolt cutter

6.2 Disassemble Preparation

Prior to Disassembling, the power must be switched off, remove the host monitor battery.

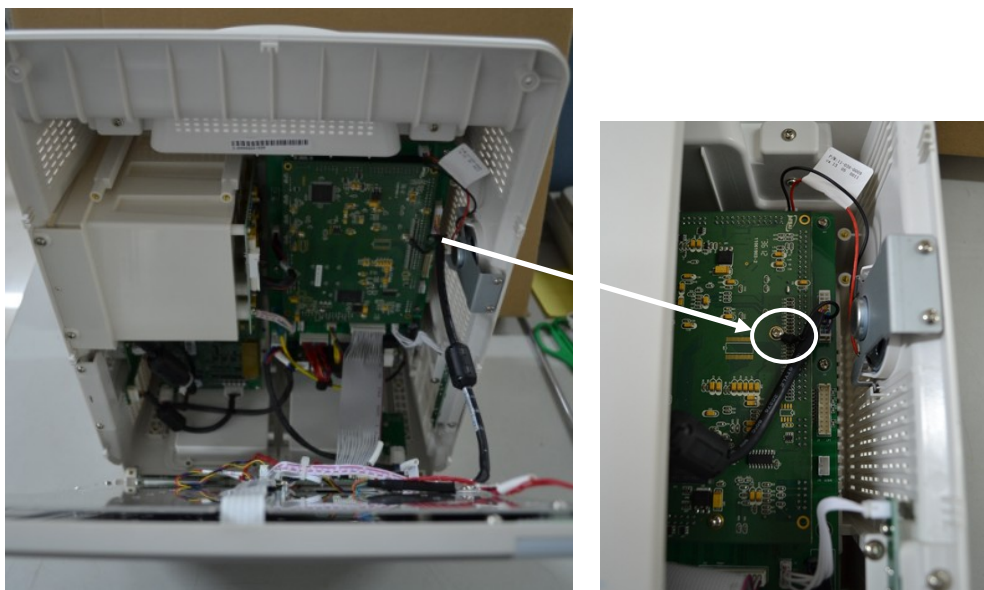
6.3 Disassemble the Host Monitor

6.3.1 Disassemble steps of Q7:

1. Unscrew the stand screws of the cover, open the monitor, then part the front and the back cover board;

Note: Before unplug the cable connecting the LCD, please unscrew the screws connecting the ground first.

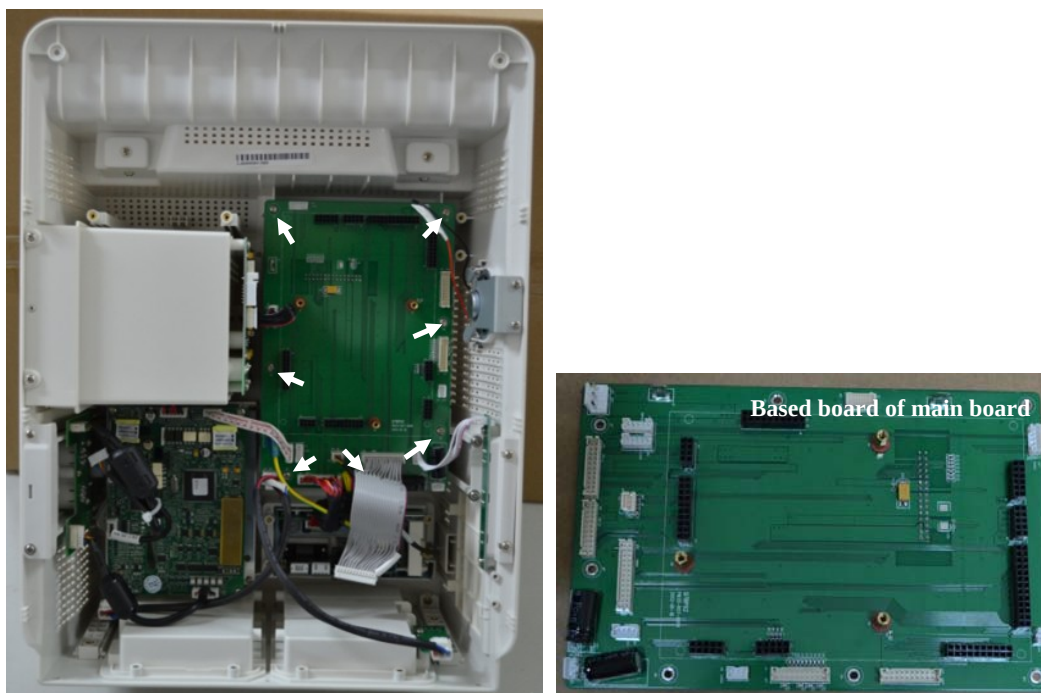




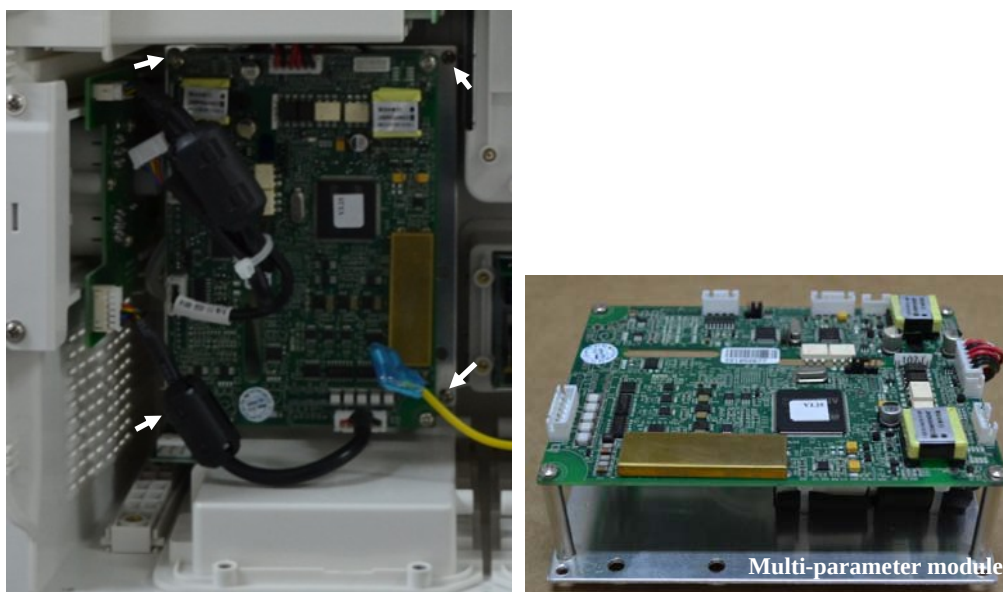
2. Disassemble the main board: Unscrew the stand screws for the main board, then get the main board out;



3. Disassemble the base board of main board: Unscrew the screws stand for the base board of main board, unplug the connecting wires, and then get the base board of main board out;



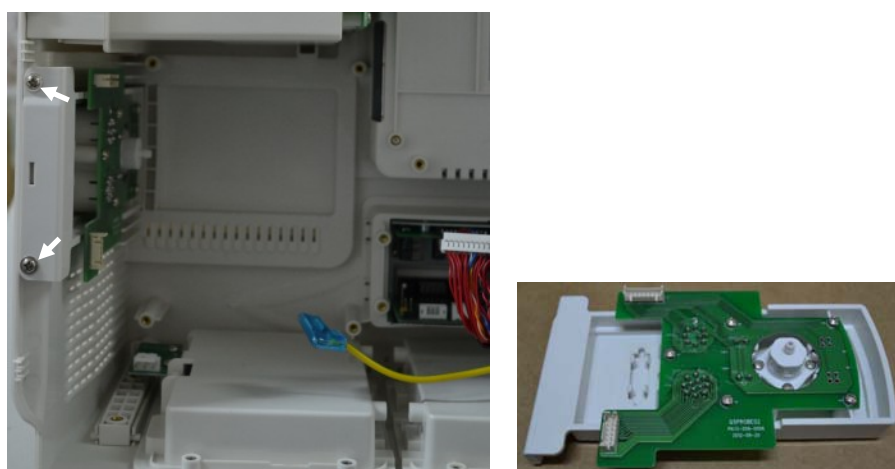
4. Disassemble multi-parameter module: Unscrew the screws stand for the multi-parameter module, unplug the connecting wires, and then get the multi-parameter module out;



5. Unplug the plug-in slot and the IO board: Unscrew the screws stand for the plug-in slot, get the plug-in slot out; Unscrew the screws stand for the IO board, get the IO board out;

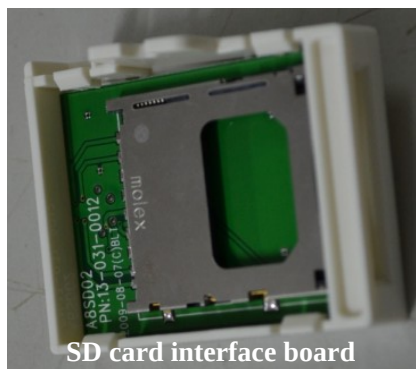
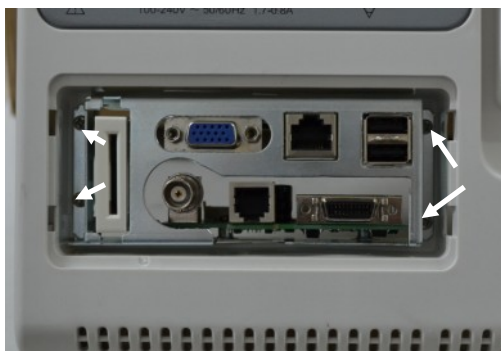


6. Disassemble the side plug-in board: Unscrew the stand screws for the side plug-in board, then get the side plug-in board out;



7. Disassemble the back plug-in board: Remove the stand cover board of back plug-in board, then unscrew the stand screws of the back plug-in board, and then get the back plug-in board out; Unscrew the relate screws, get out the SD card interface board, SOPA board and SOPB board.





SD card interface board



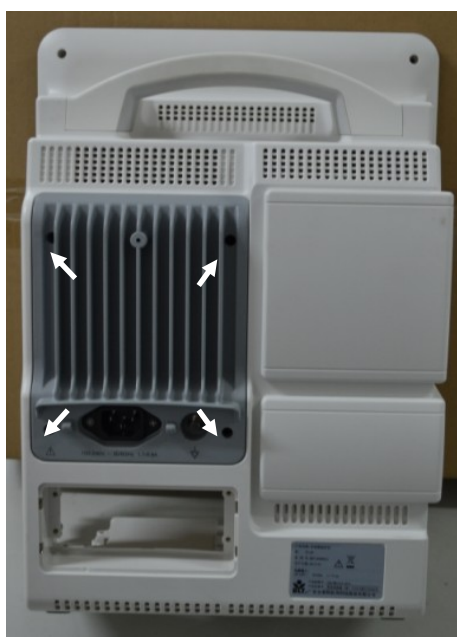
SOPA board



SOPB board



8. Disassemble the power: Unplug the power module, unscrew the stand screws of the power module, open the cover board of power, then get the power out;

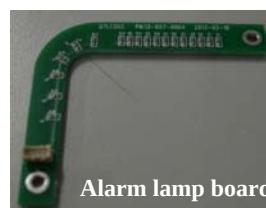




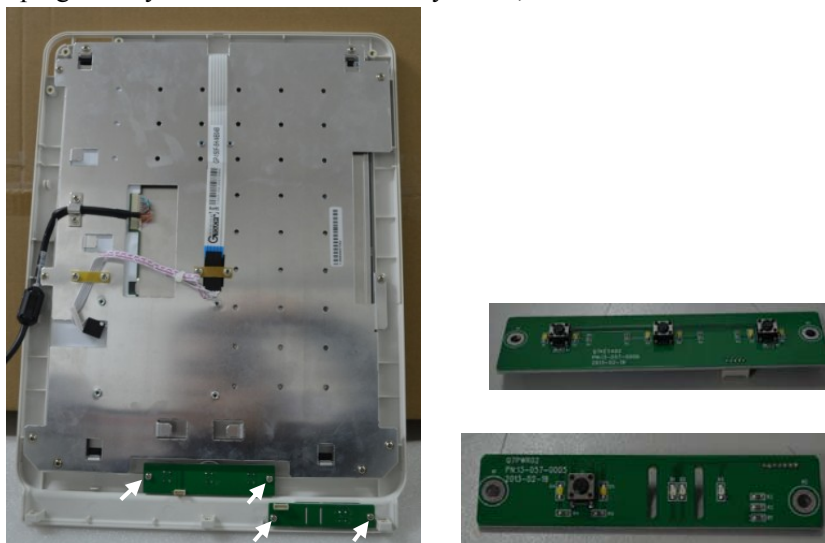
9. Disassemble the key control board: Unplug relate connecting wires, unscrew the stand screws of the key control board, then get the key control board out;



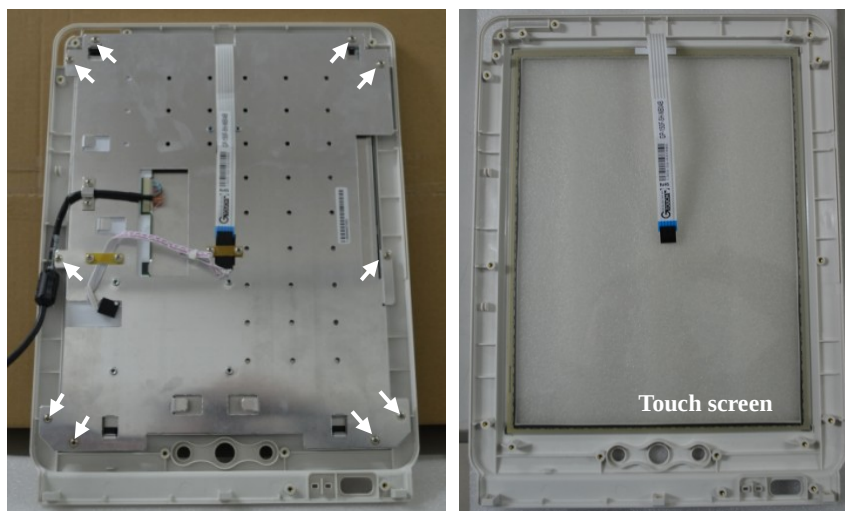
10. Disassemble the alarm lamp board: Unscrew the stand screws of the alarm lamp board, then get the alarm lamp board out;



11. Disassemble the key board: Unscrew the stand screws of the key board and the switch key board, unplug the key board and the switch key board;

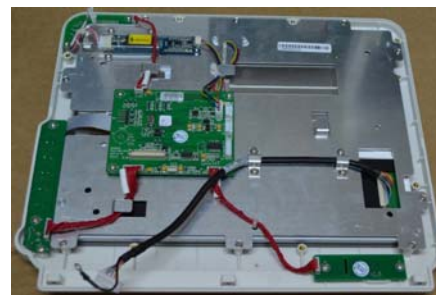


12. Disassemble the LCD: Unscrew the screws of the LCD stand, then get out the LCD board and the touch screen; Unscrew the stand screws of LCD, then get the LCD out;

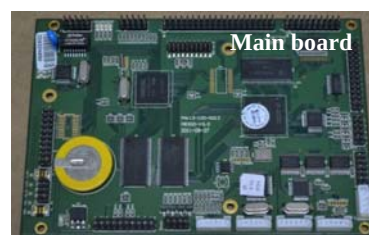


6.3.2 Disassemble steps of Q3/Q5 :

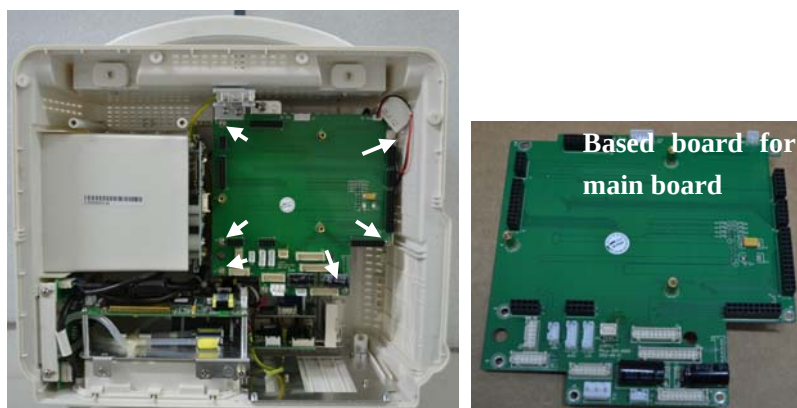
1. Unscrew the stand screws of the cover, open the monitor, then part the front and the back cover board;



4. Disassemble the main board: Unscrew the stand screws for the main board, then get the main board out;



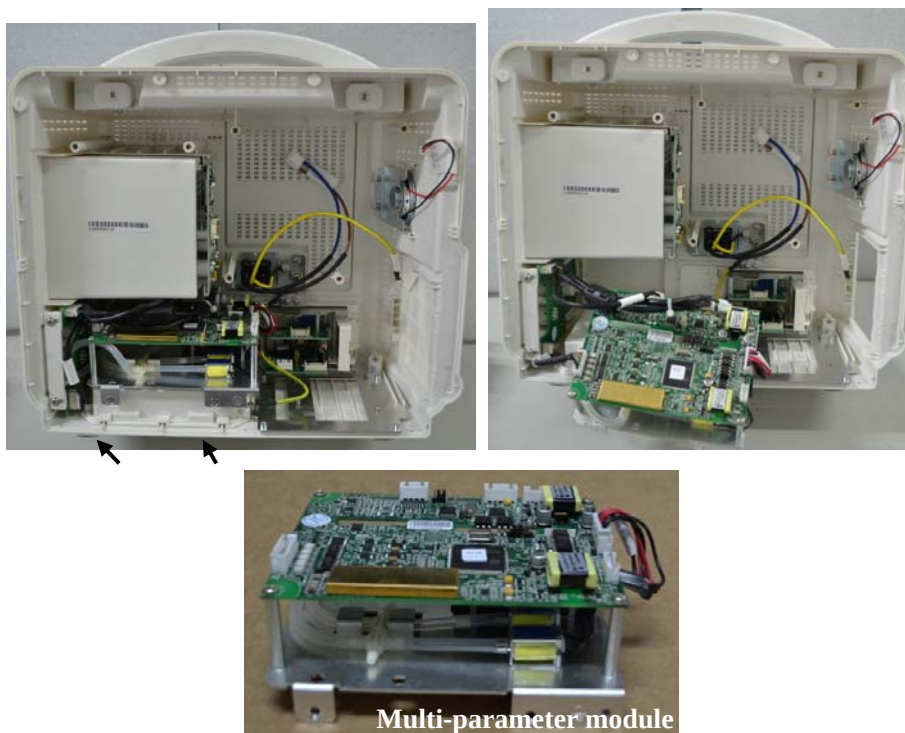
5. Disassemble the base board of main board: Unscrew the stand screws for the base board of main board, unplug the connecting wires, and then get the base board of main board out;



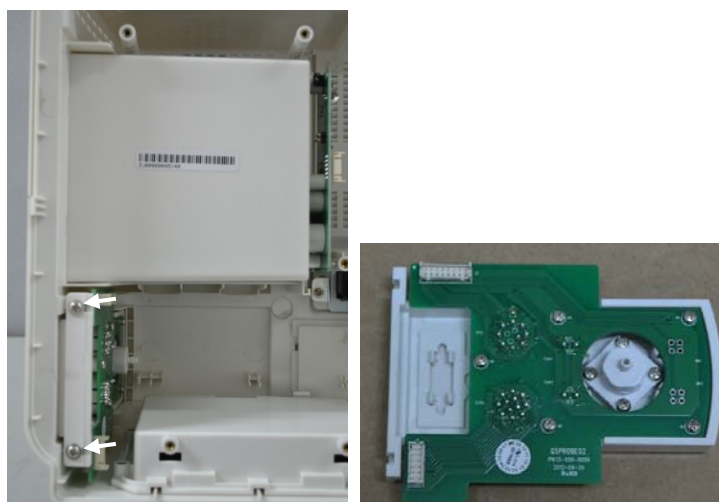
4. Disassemble the power: Unplug the power stand, get out the power and the stand, unscrew the stand screws of the power module, then get the power out;



5. Disassemble multi-parameter module: Unscrew the stand screws for the multi-parameter module, unplug the connecting wires, and then get the multi-parameter module out;



6. Disassemble the side plug-in board: Unscrew the stand screws for the side plug-in board, then get the side plug-in board out;



7. Disassemble the back plug-in board: Remove the stand cover board of back plug-in board, then unscrew the stand screws of the back plug-in board, and then get the back plug-in board out;

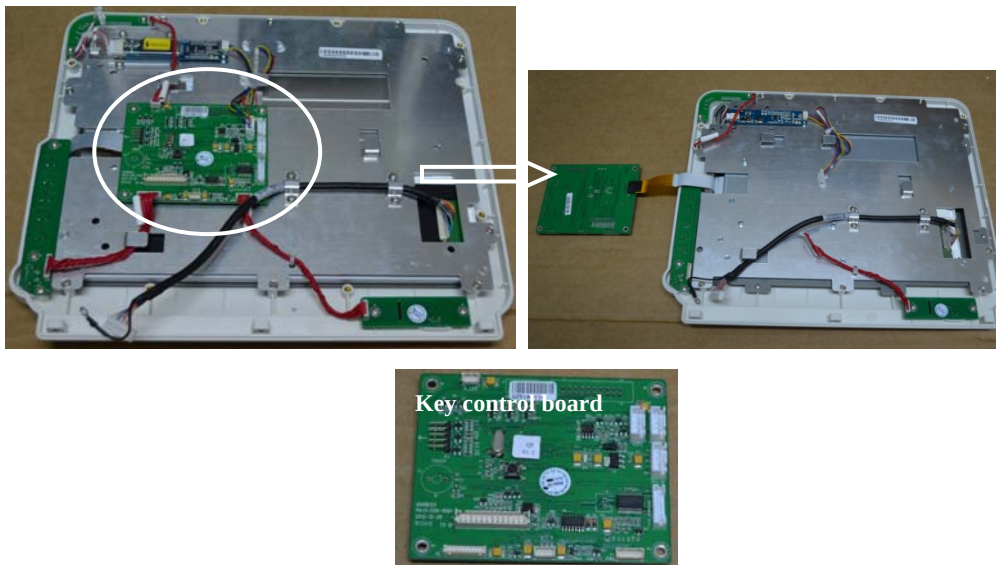




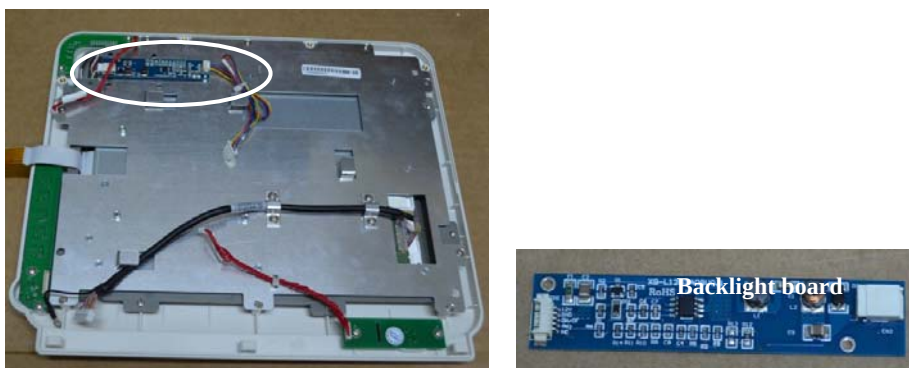
8. Disassemble the plug-in slot and IO board: Unscrew the stand screws of the plug-in slot, get out the plug-in slot; Unscrew the stand screws IO board, get out the IO board;



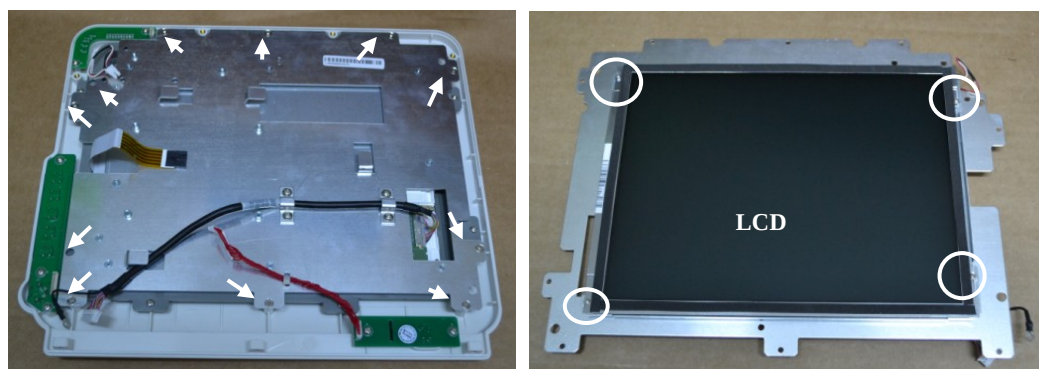
9. Disassemble the key control board: Unplug relate connecting wires, unscrew the stand screws of the key control board, then get out the key control board;

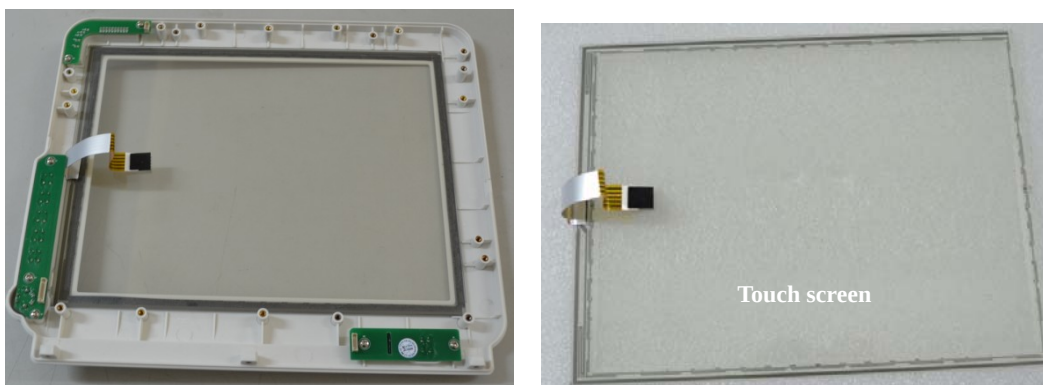


10. Disassemble the backlight board: Unscrew the stand screws of the backlight board, then get out the backlight board;

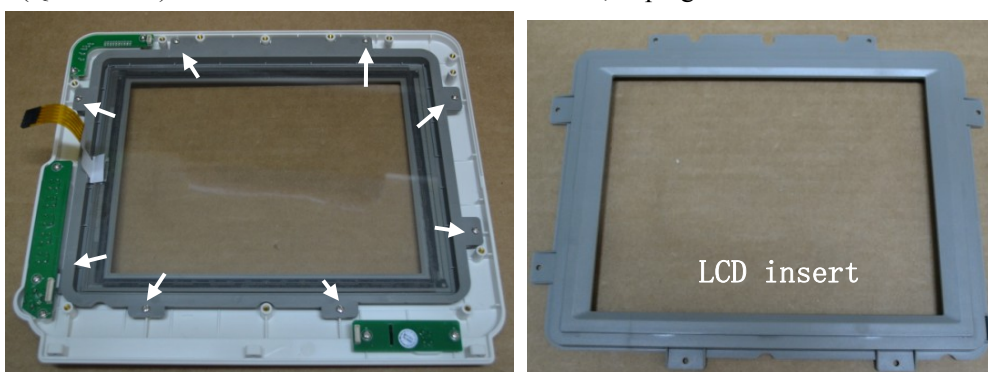


11. Disassemble the LCD: Unscrew the screws of the LCD stand, then get out the LCD board and; Unscrew the stand screws of LCD, then get the LCD out;

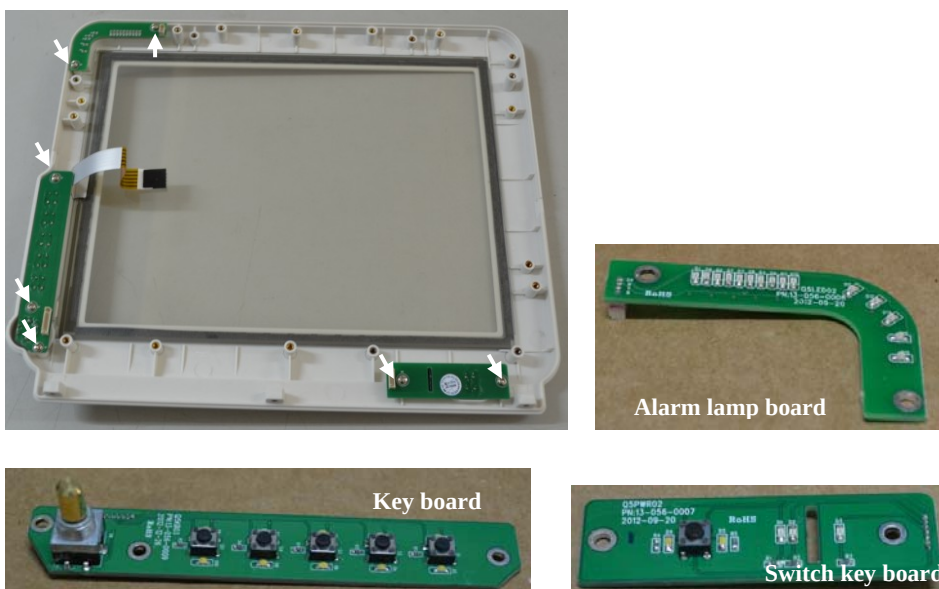




12. (Q3 monitor) Unscrew the screws of the LCD insert, unplug the LCD insert.

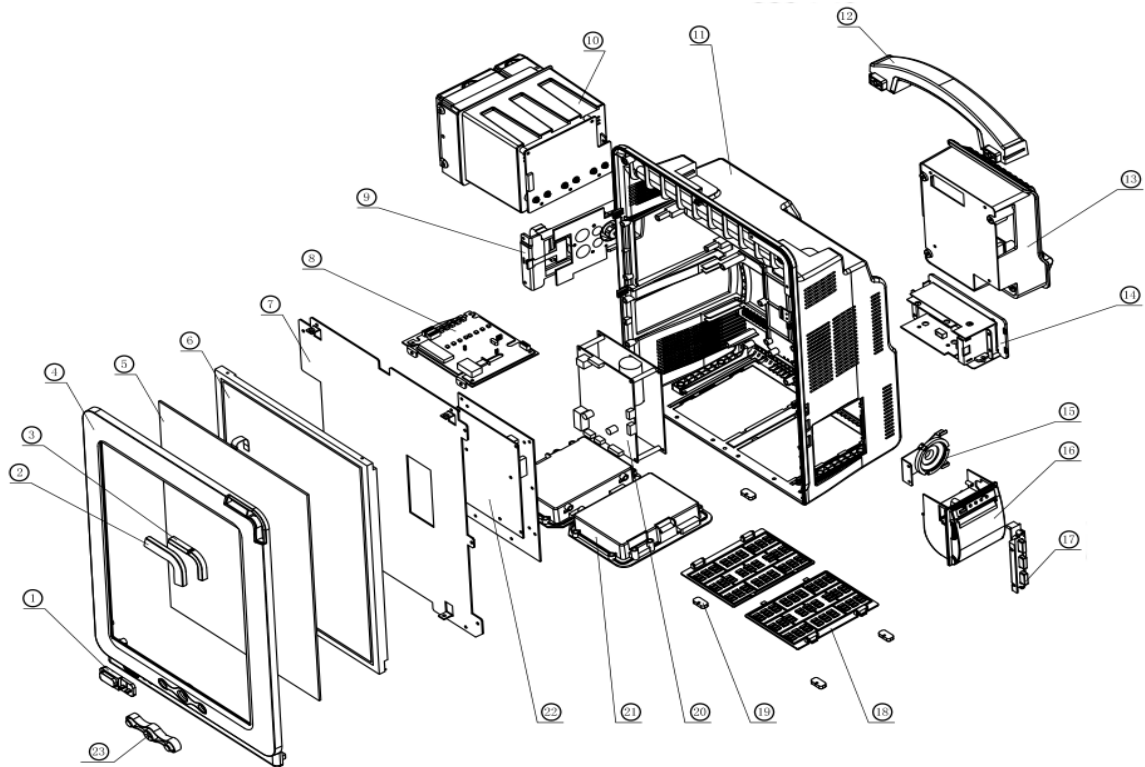


13. Disassemble the Key board: Unscrew the screws of the Key board and the Switch key board, unplug the Key board and the Switch key board.



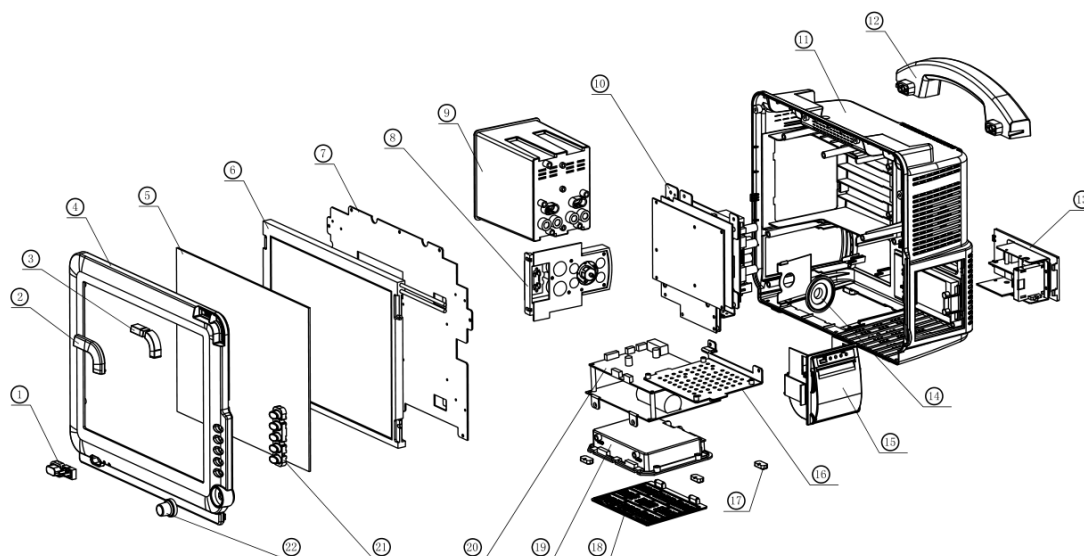
Appendix A Exploded View of Monitor

Q7:



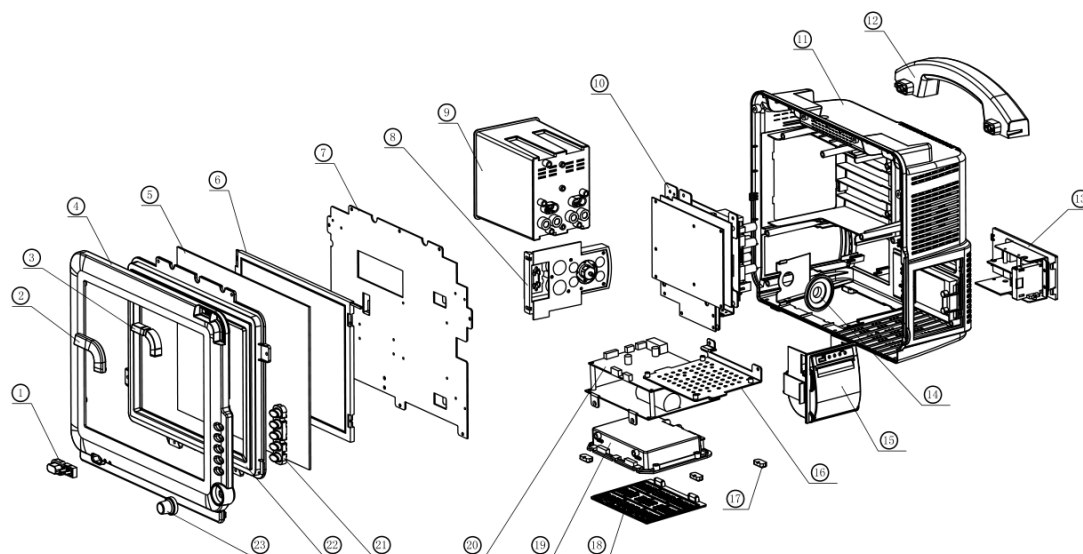
1	Power key	2	Alarm show	3	Optical filter
4	Front cover	5	Touch screen	6	LCD
7	LCD stand	8	12-lead ECG board	9	Side connect plug-in board
10	3- plug-in slot	11	Back cover	12	Handle
13	Power supply	14	Back connect plug-in board	15	Speaker
16	Recorder	17	Function keys	18	Battery cover board
19	Underlay	20	BM100A module	21	Battery case
22	Main board	23	Direction keys		

Q5:



1	Power key	2	Alarm show	3	Optical filter
4	Front cover	5	Touch screen	6	LCD
7	LCD stand	8	Side connect plug-in board	9	2- plug-in slot
10	Main board& Power module	11	Back cover	12	Handle
13	Back connect plug-in board	14	Speaker	15	Recorder
16	Power supply	17	Underlay	18	Battery cover board
19	Power case	20	BM100A module	21	Function keys
22	Trim knob				

Q3:



1	Power key	2	Alarm show	3	Optical filter
4	Front cover	5	Touch screen	6	LCD
7	LCD stand	8	Side connect plug-in board	9	2- plug-in slot
10	Main board& Power module	11	Back Cover	12	Handle
13	Back connect plug-in board	14	Speaker	15	Recorder
16	Holding hand	17	Underlay	18	Battery cover board
19	Power case	20	BM100A module	21	Function keys
22	Front cover insert	23	Trim knob		

Appendix B Products Components List

Q7 Patient monitor main unit		
No.	Materiel Code	Description
Metal		
1	1.20.39-000011-0-1	Power module cast aluminum parts
2	1.21.14-000005-005	Ground pole
3	1.20.39-000009-0-1	Battery cover board
4	1.20.57-000003-0-1	Q7 Back plug-in board
5	1.20.57-000002-0-1	Q7 Speaker fix board
6	1.20.39-000010-0-1	The stand to prevent power line from drop off
7	1.20.57-000001-0-1	Q7 LCD stand
8	1.20.37-000006-0-0	Touch screen connect line stand
Plastic cement		
9	1.19.30-000026-0-1	Plug-in case electrode stand
10	1.19.32-000018-0-1	SD Card stand
11	1.19.57-000009-0-1	Q7 Side plug-in board
12	1.19.57-000018-0-1	Q7 SpO2 plug-in cover board
13	1.19.57-000007-0-1	Q7 Handle under cover
14	1.19.57-000006-0-1	Q7 Handle upper cover
15	1.19.56-000014-0-1	Battery case
16	1.19.57-000002-0-1	Q7 Back cover
17	1.19.57-000012-0-1	Q7 Bottom stand
18	1.19.57-000010-0-1	Q7 Back connect plug-in board
19	1.19.57-000011-0-1	Q7 Key stand
20	1.19.57-000014-0-1	Q7 Function keys
21	1.19.57-000001-0-1	Q7 Front cover
22	1.19.57-000003-0-1	Q7 Lampshade
23	1.19.57-000016-0-1	Q7 Direction keys
24	1.19.57-000017-0-1	Q7 Power alarm show
25	1.19.57-000015-0-1	Q7 Power key
27	1.19.57-000013-0-1	Q7 Speaker stand
28	1.19.34-000008-1-1	Battery cover board
Boards		
29	2.31.0004-01-01	IO interface board A (with JST 10p socket)
30	2.31.0005-01-00	SD card interface board
31	1.15.01-0053-01-00	PS186 Switch power
32	2.57.0001-01-00	SOP interface board A
33	2.57.0002-01-00	SOP interface board B
34	2.56.0010-01-00	Probe interface board (2-IBP)
35	2.57.0004-01-00	Q7 bottom of main board

36	2.00.0060-09-00	MB300 Main board (VGA,V3.0,Q7 with)
37	2.57.0006-01-00	Q7 Battery interface board (right)
38	2.57.0005-01-00	Q7 Battery interface board (left)
39	2.00.0025-02-00	BM100A Multi-parameter module
40	2.57.0009-01-00	Q7 Key board A
41	2.57.0010-01-00	Q7 Key board B
42	2.57.0007-01-00	Q7 alarm show board
43	2.57.0008-01-00	Q7 Power switch key board
44	2.57.0003-01-00	Board of control Key
45	1.16.00-000080-001	15" Touch screen
46	1.14.150-0405-00-1	15" LCD
Wires		
47	1.11.23-0101004-01	Inside ground connecting wire
48	1.11.23-0103003-01	3PVH AC power connecting wire
49	1.11.31-0000011-01	Speaker connecting wire
50	1.11.29-0000009-01	SD Card interface board connecting wire
51	1.11.56-0000014-01	Q7 SpO2/Temp/IBP single connecting wire
52	1.11.58-0000004-01	Q7 5-lead ECG connecting wire
53	1.11.58-0000003-01	Q7 Battery interface board connecting wire
54	1.11.24-0105001-91	5p Switch connecting wire
55	1.11.58-0000007-01	Q7 SOP interface board A connecting wire
56	1.11.58-0000009-01	Q7 Key control board connecting wires
57	1.11.29-0000018-01	Infrared interface board communication line
58	1.11.56-0000016-01	LCD connecting wire
59	1.11.58-0000001-01	Q7 Alarm show board connecting wire
60	1.11.58-0000005-01	Q7 Key board A connecting wire
61	1.11.58-0000006-01	Q7 Key board B connecting wire
62	1.11.56-0000003-01	Power switch key board connecting wire
63	1.11.40-0000001-00	Bottom slab of the main board power wire
64	1.11.58-0000011-01	Touch screen transfer wire
65	1.11.23-0000007-01	Power line of LED back-light
66	1.11.58-0000010-01	Q7 Inside ground connecting wire
Others		
67	1.16.21-000002-001	Speaker
68	2.31.0021-01-00	NIBP connector
69	1.10.00-0032-00-01	Three-core power socket

Q5 Patient monitor main unit		
No.	Materiel Code	Description
Metal		
1	1.20.56-000001-0-1	Q5 LCD stand
2	1.20.56-000002-0-1	Q5 Main board stand
3	1.20.56-000003-0-1	Q5 Speaker stand
4	1.20.56-000004-0-1	Q5 Back plug-in board
5	1.20.56-000005-0-1	Q5 Blower stand
6	1.20.56-000006-0-1	Q5 Power socket stand board
7	1.20.56-000007-0-1	Q5 Plug-in slot stand
8	1.21.14-000005-005	Q5 Ground pole
9	1.20.41-000006-0-1	The stand to prevent power from drop off
Plastic cement		
10	1.19.56-000001-0-2	Q5 Front cover (Silk-screen Q5)
11	1.19.56-000003-0-1	Q5 Back cover
12	1.19.56-000004-0-1	Q5 Trim knob
13	1.19.56-000005-0-1	Q5 Alarm show
14	1.19.56-000008-0-1	Q5 Handle under cover
15	1.19.56-000009-0-1	Q5 Handle upper cover
16	1.19.56-000012-0-1	Q5 Side plug-in board
17	1.19.56-000013-0-1	Q5 Back plug-in cover board
18	1.19.56-000014-0-1	Q5 Battery case
19	1.19.56-000015-0-1	Q5 SpO2 socket cover board
20	1.19.56-000016-0-1	Q5 Power key
21	1.19.56-000017-0-1	Q5 Power show cover
22	1.19.56-000018-0-1	Q5 Function keys
23	1.19.32-000005-0-1	Recorder cover board
24	1.19.34-000008-1-1	Battery cover board
25	1.19.32-000018-0-1	SD Card stand
Board		
26	1.16.00-000077-002	12.1" Touch screen
27	2.56.0010-01-00	Q5 Probe interface board (2-IBP)
28	2.00.0025-03-00	BM100A Multi-parameters module(All parameters, with Q5 air pump stand)
29	2.00-0060-10-00	MB300 Main board (VGA, v3.0,Q5 with)
30	1.15.01-0047-02-01	PS139-A3 Switch power
31	1.14.121-1004-00-5	12.1" LCD
32	1.15.08-0025-01-00	Inverter (XG-L1203A08,big)
33	2.56.0001-01-00	Q5 Key control board
34	2.56.0002-01-00	SOP interface board A
35	2.56.0003-01-00	SOP interface board B
36	2.56.0004-01-00	Q5 Key board

37	2.56.0005-01-00	Q5 Bottom slab of the main board
38	2.56.0006-01-00	Q5 Battery interface board
39	2.56.0008-01-00	Q5 Alarm show board
40	2.56.0009-01-00	Q5 Power switch key board
41	2.31.0005-01-00	SD Card interface board
42	2.56.0012-00-01	Q5 2 plug-in slot groups
Wire		
43	1.11.56-0000001-01	Q5 Alarm show board connecting wire
44	1.11.56-0000002-01	Q5 Key board connecting wire
45	1.11.56-0000003-01	Q5 Power switch key board connecting wire
46	1.11.56-0000004-01	Q5 Key control board connecting wires
47	1.11.56-0000005-01	Q5 IO interface board communication line
48	1.11.56-0000006-01	Q5 SOP interface board A connecting wire
49	1.11.56-0000007-01	Q5 SOP interface board B connecting wire
50	1.11.56-0000010-01	Q5 inside ground connecting wire A
51	1.11.56-0000011-01	Q5 inside ground connecting wire B
52	1.11.56-0000012-01	Q5 inside ground connecting wire C
53	1.11.56-0000013-01	Q5 Battery interface board connecting wire
54	1.11.56-0000014-01	Q5 5-lead ECG connecting wire
55	1.11.56-0000017-01	Q5 SpO2/Temp/IBP signals connecting wire
56	1.11.56-0000018-01	Q5 LCD connecting wire
57	1.11.28-0103001-01	3PHV AC power connecting wire
58	1.11.22-0104002-01	Inverter connecting wire
59	1.11.29-0000009-01	SD card interface line
60	1.11.31-0000011-01	Speaker connecting wire
61	1.11.24-0105001-91	5p Switch connecting wire
Other		
62	1.16.21-000002-001	Speaker
63	2.31.002-01-00	NIBP connector
64	1.10.00-0033-00-02	Three-core power socket
65	1.16.18-000006-002	Blower

Q3 Patient monitor main unit		
No.	Materiel Code	Description
Metal		
1	1.20.55-000001-0-1	Q3 LCD stand
2	1.20.55-000002-0-1	Q3 LCD underlay
3	1.20.56-000002-0-1	Q5 Main board stand board
4	1.20.56-000003-0-1	Q5 Speaker stand board
5	1.20.56-000004-0-1	Q5 Back plug-in board
6	1.20.56-000005-0-1	Q5 Blower stand board
7	1.20.56-000006-0-1	Q5 Power socket stand board
8	1.20.56-000007-0-1	Q5 Stand of plug-in slot
9	1.21.14-000005-005	Q5 Ground pole
10	1.20.41-000006-0-1	The stand to prevent power to drop off
Plastic cement		
11	1.19.56-000001-0-2	Q5 Front cover (Silk-screen Q3)
12	1.19.56-000002-0-1	Q5 Front cover inserts
13	1.19.56-000003-0-1	Q5 Back cover
14	1.19.56-000004-0-1	Q5 Trim knob
15	1.19.56-000005-0-1	Q5 Alarm show
16	1.19.56-000008-0-1	Q5 Handle under cover
17	1.19.56-000009-0-1	Q5 Handle upper cover
18	1.19.56-000012-0-1	Q5 side plug-in board
19	1.19.56-000013-0-1	Q5 Back connect plug-in cover board
20	1.19.56-000014-0-1	Q5 Battery case
21	1.19.56-000015-0-1	Q5 SpO2 socket cover board
22	1.19.56-000016-0-1	Q5 Power key
23	1.19.56-000017-0-1	Q5 Power show cover
24	1.19.56-000018-0-1	Q5 Function keys
25	1.19.32-000005-0-1	Recorder cover
26	1.19.34-000008-1-1	Battery cover
27	1.19.32-000018-0-1	SD card stand
Board		
28	1.16.00-000074-001	10.4' Touch screen
29	2.56.0010-01-00	Q5 Probe interface board(2-IBP)
30	2.00.0025-03-00	BM100A Multi-parameter module (All parameters, with Q5 air pump stand)
31	2.00-0060-10-00	MB300 Main board (VGA, v3.0,Q5 with)
32	1.15.01-0047-02-01	PS139-A3 Switch power
33	1.14.104-1704-00-1	10.4" LCD
34	1.15.08-0025-01-00	Inverter (XG-L1203A08, Big)
35	2.56.0001-01-00	Q5 Key control board
36	2.56.0002-01-00	SOP Interface board A

37	2.56.0003-01-00	SOP Interface board B
38	2.56.0004-01-00	Q5 Key board
39	2.56.0005-01-00	Q5 Bottom slab of the main board
40	2.56.0006-01-00	Q5 Battery interface board
41	2.56.0008-01-00	Q5 Alarm show board
42	2.56.0009-01-00	Q5 Power switch key board
43	2.31.0005-01-00	SD card interface board
44	2.56.0012-00-01	Q5 2 plug-in slot groups
Wire		
45	1.11.56-0000001-01	Q5 Alarm show board connecting wire
46	1.11.56-0000002-01	Q5 Key board connecting wire
47	1.11.56-0000003-01	Q5 Power switch key board connecting wire
48	1.11.56-0000004-01	Q5 Key control board connecting wires
49	1.11.56-0000005-01	Q5 IO interface board communication line
50	1.11.56-0000006-01	Q5 SOP interface board A connecting wire
51	1.11.56-0000007-01	Q5 SOP interface board B connecting wire
52	1.11.56-0000010-01	Q5 inside ground connecting wire A
53	1.11.56-0000011-01	Q5 inside ground connecting wire B
54	1.11.56-0000012-01	Q5 inside ground connecting wire C
55	1.11.56-0000013-01	Q5 Battery interface board connecting wire
56	1.11.56-0000014-01	Q5 5-lead ECG connecting wire
57	1.11.56-0000017-01	Q5 SpO2/Temp/IBP signals connecting wire
58	1.11.56-0000018-01	Q5 LCD connecting wire
59	1.11.28-0103001-01	3PHV AC power wire
60	1.11.22-0104002-01	Inverter connecting wire
61	1.11.29-0000009-01	SD card interface board connecting board
62	1.11.31-0000011-01	Speaker connecting wire
63	1.11.24-0105001-91	5p Switch connecting line
Other		
64	1.16.21-000002-001	Speaker
65	2.31.0021-01-00	NIBP connector
66	1.10.00-0033-00-02	Three-core power socket
67	1.16.18-000006-002	Blower

Product Model: Q3/Q5/Q7 Patient monitor

Product Name: Patient monitor

Manufacturer Name: Guangdong Biolight Meditech Co.,Ltd.

Address: No.2 Innovation First Road, Technical Innovation Coast, Hi-tech Zone, Zhuhai,
P.R.China

Post code: 519085

PN: 22-048-0008